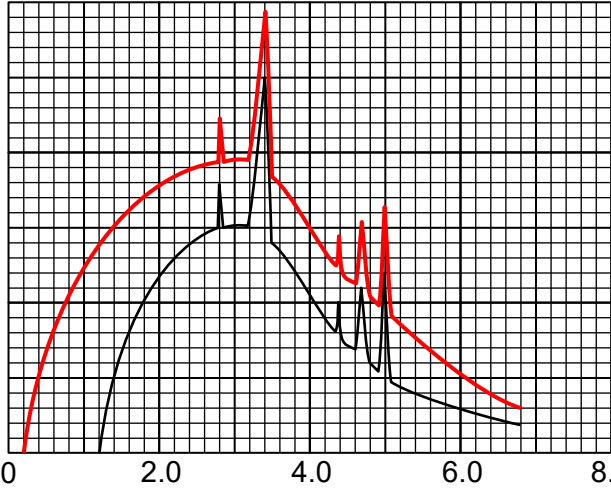


		power.
7(a)(i)	During de-excitation, the photons are <u>emitted uniformly in all directions</u> . Hence the intensity of the radiation in the direction of the incident photons is a <u>small fraction of the original intensity</u> .	[1]

No.	Solution	Remarks
7(a)(ii)	$\Delta E = \frac{hc}{\lambda}$ $\lambda_{12} = \frac{hc}{\Delta E_{12}}$ $\lambda_{12} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{[(-3.04) - (-5.14)] \times (1.60 \times 10^{-19})}$ $= 592 \text{ nm (visible)}$ $\lambda_{13} = \frac{hc}{\Delta E_{13}}$ $\lambda_{13} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{[(-1.53) - (-5.14)] \times (1.60 \times 10^{-19})}$ $= 344 \text{ nm (invisible)}$ The wavelength of light corresponding to transition from E_1 to E_3 is outside the visible light range, Hence, only one absorption spectral line, corresponding to the E_1 to E_2 transition, is observed. OR $\Delta E_{\max} = \frac{hc}{\lambda_{\min}}$ $= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{400 \times 10^{-9}}$ $= 4.9725 \times 10^{-19} \text{ J}$ $= 3.108 \text{ eV}$ Expected energy level = $-5.14 + 3.108 = -2.032 \text{ eV}$ $-2.032 \text{ eV} < -1.53 \text{ eV}$ (expected energy level does not exist) Since only level E_2 is attainable from level E_1, only one absorption spectral line is observed.	[1] [1] [1] [1] [1] [1]
7(b)(i)	When the highly energetic electrons strike the metal target and are suddenly (or rapidly) decelerated by collision with the target metal, some electrons lose all their energy in a single interaction (or event). The energy lost is emitted as X-ray photons of equivalent maximum energy.	[1] [1]

No.	Solution	Remarks
7(b)(ii)	$\lambda \approx 3.4 \times 10^{-11} \text{ m}$ $p = \frac{h}{\lambda}$ $= \frac{(6.63 \times 10^{-34})}{3.4 \times 10^{-11}}$ $= 1.95 \times 10^{-23} \text{ kg m s}^{-1}$	[1]
7(b)(iii)	relative Intensity  wavelength / 10^{-11} m	[1] - same characteristic wavelengths, - lower cut-off wavelength, - higher overall intensity - continuous spectrum should be smooth.